Experiment 5 – SQL Conditional Logic (Odd & Even Values)

Experiment

Experiment 5: SQL Conditional Logic using the MOD (%) operator to classify employee salary values as Odd or Even. This experiment demonstrates numerical analysis and conditional evaluation using SQL queries in Oracle and PostgreSQL.

Aim

To understand and apply conditional logic in SQL by using the modulus operator (MOD / %) to analyze numerical data and classify employee salaries as odd or even, thereby improving data analysis and decision-making skills in SQL.

Objective

- To write and execute SQL queries using the MOD (%) operator.
- To retrieve salary details from an employee table.
- To classify salaries into Odd and Even categories.
- To display odd and even salary values separately.

Software Requirements

Database Management System:

- Oracle Database Express Edition (Oracle XE)
- PostgreSQL Database

Database Administration Tool / Client Tool:

- Oracle SQL Developer (for Oracle XE)
- pgAdmin (for PostgreSQL)

Code:

```
CREATE TABLE employee (
  id INT PRIMARY KEY,
  name VARCHAR(50),
```

```
  department VARCHAR(50),
  salary INT
);

INSERT INTO employee VALUES (1, 'Rahul', 'HR', 25001);
INSERT INTO employee VALUES (2, 'Sneha', 'Finance', 32000);
INSERT INTO employee VALUES (3, 'Amit', 'IT', 45015);
INSERT INTO employee VALUES (4, 'Neha', 'Marketing', 28000);
INSERT INTO employee VALUES (5, 'Vikram', 'Sales', 39011);

SELECT id, name, department, salary
FROM employee;

SELECT
  name,
  salary,
  CASE
    WHEN salary % 2 = 0 THEN 'Even Salary'
    ELSE 'Odd Salary'
  END AS salary_type
FROM employee;

SELECT
  name,
  salary
FROM employee
```

```
WHERE salary % 2 <> 0;


SELECT
  name,
  salary
FROM employee
WHERE salary % 2 = 0;
```

Practical / Experiment Steps

1. Create an employee table with columns:
    a. id
    b. name
    c. department
    d. salary
2. Insert sample employee salary records into the table.
3. Write a SQL query to retrieve employee salary details.
4. Use the MOD(salary, 2) function (Oracle) or salary % 2 (PostgreSQL) to check whether the salary is odd or even.
5. Use a CASE statement to classify salaries as:
    a. 'Odd Salary'
    b. 'Even Salary'
6. Execute separate queries to:
    a. Display only employees with Odd salaries.
    b. Display only employees with Even salaries.
7. Observe and record the output results.

Procedure

1. Open Oracle SQL Developer or pgAdmin.
2. Connect to the database.
3. Create the employee table using CREATE TABLE.
4. Insert sample records using INSERT INTO.
5. Execute SELECT queries with MOD operator.

6. Apply CASE statement for conditional classification.
7. Display results and capture screenshots.

Input / Output Details

Input

Table Name: employee

Columns:

- id (Number)
- name (Varchar)
- department (Varchar)
- salary (Number)

Logic Used:

- MOD(salary, 2)
- CASE statement

Step-wise Output

Step 1 – Create employee Table

Step 2 – Insert Data into employee Table

Step 3 – Display Employee Salary Records

Step 4 – Classify Salaries as Odd or Even

Step 5 – Display Only Odd Salaries

Step 6 – Display Only Even Salaries

SCREENSHOT:

Output:

```
+----+--------+------------+--------+
| id | name   | department | salary |
+----+--------+------------+--------+
|  1 | Rahul  | HR         |  25001 |
|  2 | Sneha  | Finance    |  32000 |
|  3 | Amit   | IT         |  45015 |
|  4 | Neha   | Marketing  |  28000 |
|  5 | Vikram | Sales      |  39011 |
+----+--------+------------+--------+
+--------+--------+-------------+
| name   | salary | salary_type |
+--------+--------+-------------+
| Rahul  |  25001 | Odd Salary  |
| Sneha  |  32000 | Even Salary |
| Amit   |  45015 | Odd Salary  |
| Neha   |  28000 | Even Salary |
| Vikram |  39011 | Odd Salary  |
+--------+--------+-------------+
+--------+--------+
| name   | salary |
+--------+--------+
| Rahul  |  25001 |
| Amit   |  45015 |
| Vikram |  39011 |
+--------+--------+
+-------+--------+
| name  | salary |
+-------+--------+
| Sneha |  32000 |
| Neha  |  28000 |
```

Learning Outcome

After completing this experiment, the learner will be able to:

- Understand conditional logic in SQL queries.
- Use the MOD (%) operator for numerical analysis.
- Classify numeric values into odd and even categories.
- Write efficient SQL queries for payroll data analysis.

- Apply SQL logic in real-world enterprise environments.